

SIH26011 — Master Research & Implementation Specification

Project: 3D ULPIN Generation and Vertical Property Mapping System
Problem ID: SIH26011
Organisation: Ministry of Rural Development — Department of Land Resources (DoLR)
Category / theme: Software · Smart Automation
Official source: sih.gov.in/sih2026PS (verified 18 September 2026)
Document status: Implementation-ready engineering specification
Rule: This document separates **verified facts** from **our engineering proposal**. A proposed identifier is never called official ULPIN.

1. Executive summary

India already has a national surface-parcel identifier: **ULPIN / Bhu-Aadhaar**, a 14-character alphanumeric ID generated from geo-referenced parcel vertices under DILRMP. DoLR states that this unique ID “would spatially be pointing to the **surface** of the parcel.” That is why 2D ULPIN cannot uniquely identify a fifth-floor flat, a basement parking volume, an air-rights slab, or a sewer easement under the same plot.

Fact: No official Government of India 3D ULPIN standard was found in DoLR, DILRMP, or related public documentation as of 18 September 2026. SIH26011 is therefore an **open design problem**, not a request to implement a published 3D code format.

Our 3D ULPIN is a proposed extension/representation compatible with the existing ULPIN concept.

The product we will actually finish is **BhuvanRekha-3D** (working name: **ULPIN-3D**): a demonstrable 3D cadastral engine that:

1. Ingests a 2D cadastral polygon plus optional LiDAR/DSM, floor plans, and GNSS metadata.
2. Extracts or accepts a building footprint and height.
3. Segments floors and constructs **legal 3D spatial units** (apartment, common, parking, underground corridor).
4. Validates topology deterministically (not with a neural net).
5. Issues a **proposed 3D ULPIN** that always references the parent surface ULPIN.
6. Stores records in a **LADM-aligned** PostGIS model and visualises them in Cesium so a non-technical judge understands the story in 30 seconds.

What we will not do: train PointNet from scratch; claim government LiDAR we do not have; store CityGML as the system of record; encode full geometry inside the identifier; present AI geometry as legal title.

Real novelty (defensible): a **ULPIN-compatible vertical identity + legal-space engine** that separates physical geometry from rights, with deterministic topology validation and provenance. That combination is what international 3D cadastre research treats as the hard problem, and what Indian 2D ULPIN does not currently provide.

2. Problem interpretation

2.1 Official problem statement (verbatim structure)

Verified on the official SIH 2026 portal (<https://www.sih.gov.in/sih2026PS>) on 18 September 2026:

Field	Value
PS Number	SIH26011
Title	3D ULPIN Generation and vertical Property Mapping SYstem
Organisation	Ministry of Rural Development
Department	Dept of land resources (DoLR)
Category	Software

Field	Value
Theme	Smart Automation
Idea deadline	30 September 2026
Submitted ideas (that date)	15/500
Dataset link	empty
YouTube / contact	empty

Background (official): conventional 2D land-record systems cannot uniquely define ownership rights for multi-storey apartments, underground infrastructure, elevated transport corridors, parking spaces, air-rights, and subsurface utility networks.

Required identities: surface land parcels · multi-storey apartments · underground infrastructure.

Required integrations: drone imagery · LiDAR/3D point clouds · GIS parcel layers · building floor plans · GNSS/CORS coordinates · DEM/DSM.

Named AI/ML capabilities: automated building extraction · floor segmentation · vertical parcel delineation · intelligent topology validation.

Expected outcome: standardised 3D ULPINs · mapping of vertical and underground ownership rights · volumetric cadastre · urban property governance · fewer ownership conflicts · better infrastructure/utility planning.

Important: the portal provides **no dataset**. Any claim that “SIH gave us Indian LiDAR” is false.

2.2 What the problem is really asking

The shopping list is a **capability map**, not a mandate to build a national production cadastre in a hackathon. The scientifically honest reading:

1. **Identifier problem** — extend ULPIN from 2D surface lots to 3D legal spaces without breaking the existing 14-character parent.
2. **Geometry problem** — turn 2D footprints + heights + floor plans into closed volumetric parcels.
3. **Legal-space problem** — distinguish apartment units, common areas, parking, air rights, and utility corridors.
4. **Validation problem** — detect overlapping/gapped/invalid 3D parcels.
5. **Interop problem** — talk to GIS, GNSS, and (later) government land-record systems.

2.3 Problem class map (do not treat everything as AI)

Capability	Class	MVP method
Surface parcel identification	Cadastral / geospatial	Ingest official polygon + official ULPIN if present
Multi-storey apartment ID	Cadastral + geometric	Floor-plan extrusion into legal units
Underground mapping	Geospatial + geometric	Centerline + buffer solid
3D ULPIN generation	Identifier design	Hierarchical proposed ID; UUID persistence
Drone imagery	Computer vision / photogrammetry	Optional orthomosaic; not required for core path
LiDAR / point cloud	Geospatial + geometry	PDAL + Open3D classical extraction

Capability	Class	MVP method
GIS parcel layer	Geospatial	GeoJSON/shapefile → PostGIS
Floor plans	Geometric + cadastral	Digitised polygons / GeoJSON; IFC later
GNSS/CORS	Geospatial	CRS metadata + accuracy class; not a live CORS client
DEM/DSM	Geospatial	nDSM = DSM – DTM for height
Building extraction	Geometry (+ optional ML later)	CSF + nDSM + planarity
Floor segmentation	Geometry + fusion	Z-histogram + RANSAC + plan overlay
Vertical parcel delineation	Geometric	Extrude polygon to [zmin, zmax] solid
Topology validation	Deterministic geometry	PostGIS/SFCGAL/Shapely — not ML
Volumetric cadastre	Data model	LADM-aligned spatial units
Ownership ambiguity reduction	Legal data model	RRR tables; common-area as shared right
Utility planning	Geospatial	Underground spatial units + conflict checks

Rejected interpretation: “we must ship a trained deep-learning stack for every named AI item.” Topology validation is a computational-geometry problem. Vertical delineation is extrusion + solid construction. Building extraction and floor detection can be classical for MVP and still satisfy the PS if we expose them as automated processing stages with confidence scores.

2.4 Indian legal overlay we must respect (not copy foreign law)

India does **not** have a national 3D cadastre statute. What exists:

- **ULPIN** identifies the **land parcel**, not the apartment.
- **State Apartment Ownership Acts** (e.g. Karnataka 1972, Punjab 1995, Tamil Nadu 2022) treat an apartment plus an **undivided share of common areas** as heritable immovable property. Common areas cannot be partitioned from the apartment.
- **RERA 2016 s.17** requires conveyance of the apartment to the allottee and undivided proportionate title in common areas to the association of allottees.

Transferable international concepts (technical, not legal copy):

Concept	Source	Use in India prototype
Legal space ≠ physical building	FIG 3D Cadastres Best Practices; ISO 19152	Separate <code>spatial_unit</code> from <code>building mesh</code>
Building-format vs volumetric-format lots	Queensland Land Title Act / Registrar Directions	Apartments = building-bounded; utilities/tunnels = volumetric
3D PDF / 3D certificate as first operational step	Netherlands Kadaster (Delft 2016); Shenzhen 3D certificates	Demo “cadastral record card”, not a legal title
Strata / airspace / subterranean lots	Singapore SLA	Types we model: UNIT, COMMON, AIR, SUBSURFACE
Undivided common property	Indian apartment acts + RERA	<code>COMMON</code> spatial unit + shared RRR, not fake exclusive titles

Not transferable: Queensland's statutory volumetric plan rules; Dutch deed law; China's 3D land-use right certificates. We model **spaces and rights**, we do not issue Indian titles.

3. Existing ULPIN research (facts)

3.1 What ULPIN is

DoLR official page *Bhu-Aadhar* : *Unique Land Parcel Identification Number (ULPIN)* (dolr.gov.in/en/ulpin/, last updated 25 April 2024):

Unique Land Parcel Identification Number (ULPIN) is part of DILRMP. It is a **14-digit identification number** accorded to a land parcel based on the **longitude and latitude coordinates** of the land parcel and depends on detailed surveys and geo-referenced cadastral maps.

It is described as a “Single, Authoritative Source of Truth” for a parcel, intended to support integrated land services. DoLR says the system complies with **ECCMA** and **OGC** standards.

DILRMP guidelines PDF (DoLR, April 2024) repeats: ULPIN is for generating and assigning **14 digits — alphanumeric unique ID** for each land parcel based on **geo-coordinates of vertices**.

3.2 How it is generated (what is public)

DoLR generation notes:

1. There is a formula to generate ECCMA-prescribed 14-digit Unique ID **Property Natural Identifier Unit (PNIU)** using the parcel's geo-referenced **vertices**.
2. This ID is organically dependent on parcel vertices in lat/long as **Property Natural Identifier Lot (PNIL)**, and the Unique ID (PNIU) would **spatially be pointing to the surface of the parcel**.

A 16 January 2020 DoLR presentation (circulated copies, e.g. MediaNama archive of *UNIQUE LAND PARCEL IDENTIFICATION NUMBER (ULPIN)*) adds:

- Implementation via webservice: **input = parcel geometry + EPSG code; output = PNIU, PNIL**.
- NIC / Survey of India / DST / NRSC to provide technical support.
- Some states historically used **different** local codes (AP random state-prefixed codes; UP 16-digit village+khasra codes). ULPIN is meant to standardise over those.

Uncertainty (flagged): the exact public algorithm that maps a vertex ring to the 14-character ULPIN is **not published as open source**. We will **ingest** an official ULPIN when provided. We will **not** claim that a hash we invent is a government-issued ULPIN.

3.3 ECCMA / ISO relationship (important and easy to get wrong)

ECCMA's natural-location work is now reflected in **ISO 8000-118:2025** (Natural Location Identifiers). Public summaries of that standard (and ECCMA's eNLI site) describe a reversible encoding of **WGS84 latitude, longitude, and storey or elevation** into a compact alphanumeric string (14 characters with storey, 15 with elevation). PNIU was originally described as applying a reversible algorithm to the **position and elevation of the midpoint of the plane representing the primary point of entry** to a legally delineated unit space.

That means:

- ECCMA/ISO NLI is a **point-based location code** (entrance / coordinate + floor).
- DoLR ULPIN, as implemented, is described as a **surface-parcel** ID derived from **vertices**, pointing at the **surface**.
- A floor-encoded NLI is **not** a volumetric cadastral parcel ID. Two flats can share an entrance corridor. A sewer has no “storey”. A parcel mutation changes vertices and therefore can change a geometry-derived ID.

Decision: use ISO 8000-118-style NLI, if implemented, only as an optional **location index**, never as the cadastral 3D ULPIN.

3.4 Coverage and ownership

- Rolled out in **29 States/UTs** (DoLR list includes AP, Jharkhand, Goa, Bihar, Odisha, Sikkim, Gujarat, Maharashtra, Rajasthan, Haryana, Tripura, Chhattisgarh, J&K, Assam, MP, Nagaland, Mizoram, Tamil Nadu, Punjab, DNH&DD, HP,

West Bengal, UP, Uttarakhand, Kerala, Ladakh, Chandigarh, Karnataka, NCT of Delhi).

- Pilot in 4: Puducherry, Telangana, Manipur, A&N.
- Some states use ULPIN inside **SVAMITVA**.
- DoLR says ULPIN **may include** ownership details besides size and lat/long. Ownership is **linked**, not encoded as the ID itself. The ID is a key; RoR remains the rights register.

3.5 Does official 3D ULPIN exist?

No public official 3D ULPIN standard was found. Searches of DoLR ULPIN pages, DILRMP guidelines, National Geospatial Policy 2022, and Survey of India CORS/geodetic pages did not yield a 3D ULPIN schema, 3D PNIU profile, or gazette notification for vertical identifiers.

National Geospatial Policy 2022 (DST notification S.O. 6095(E), 28 December 2022) **does** list Land Parcels as a National Fundamental Geospatial Data Theme (DoLR rural / MoHUA urban) and calls for a strategy to map **subsurface infrastructure in cities in 3D**. That is policy intent, not a 3D ULPIN spec.

Conclusion: 3D ULPIN is currently an **open design problem**. SIH26011 is the DoLR asking teams to propose a workable extension.

3.6 Compatibility rule we will implement

Official surface ULPIN (14-char, if known)

-
- proposed 3D ULPIN for building envelope
- proposed 3D ULPIN for floor slab (optional)
- proposed 3D ULPIN for unit / common / parking
- proposed 3D ULPIN for underground / airspace volumes

Compatibility means:

1. Never overwrite or “improve” an official 14-character ULPIN.
2. Every 3D unit stores `parent_ulpin` (official or UNKNOWN + proposed surface key).
3. Geometry lives in the database, not inside the identifier.
4. Display strings are clearly labelled **PROPOSED**.

4. 3D cadastre research

4.1 Core definition

A 3D cadastre registers **rights, restrictions and responsibilities (RRR)** against **3D spatial units** (legal spaces), not merely against a 2D polygon. FIG *Best Practices 3D Cadastres* emphasises that the object of registration is the **legal space**, which may coincide with, sit inside, or cut through physical buildings and terrain.

4.2 Jurisdiction comparison (what to steal technically)

Jurisdiction	Status	Useful idea	Do not copy
Netherlands	Operational 3D visualisation in deeds (Delft railway station, 2016); full 3D parcel DB still evolving	BIM → legal volume → registrable 3D view; start inside existing legal frames	Dutch deed law
Queensland, AU	Building Format Plans (walls/floors/ceilings) and Volumetric Format Plans (3D-located bounding surfaces)	Two geometry families: apartment vs tunnel/utility	Land Title Act procedures

Jurisdiction	Status	Useful idea	Do not copy
Singapore	Strata, airspace, subterranean lots exist; many plans still 2D with heights	Lot types we need; coordinated cadastre	SLA survey regulations
Shenzhen, CN	Long-running 3D cadastre practice; 2025 3D real-estate certificate (Qianhai fire station)	3D certificate + QR to 3D model; topology checks in land supply	Chinese land-use right system
Spain / others	Apartment 3D solutions	Unit-level visualisation	Not a full multi-level cadastre

4.3 Indian apartment / common / parking / air / underground

For the prototype we adopt **legal-space types**, mapped to Indian practice in comments, not as a claim of statutory 3D lots:

U class	Meaning	Typical Indian analogue
PARCEL	Surface cadastral lot	ULPIN parcel / survey number
BUILDING	Physical envelope (not a title)	Structure on the parcel
FLOOR	Storey slab (administrative)	Floor in sanctioned plan
UNIT	Exclusive apartment/office	Apartment under state AOA + RERA
COMMON	Corridor, lift, stair, lobby	Undivided common areas → association
PARKING	Exclusive or limited-common parking	Deed / RERA allotment
BALCONY	Private or limited-common projection	Per sanctioned plan
AIR	Volume above a defined height	Air rights (rarely titled today)
SUBSURFACE	Underground parking / cellar	Basement as part of building or separate
UTILITY	Pipe/cable/tunnel corridor	Easement / utility ROW (often not cadastral)

Mandatory split: `geom_physical` (walls, pipes, DSM) vs `geom_legal` (the space the right applies to). A wall is physical; the median plane or interior face used as a boundary is a legal-space modelling choice. MVP uses **interior of floor-plan polygons extruded between storey z-values** as legal space. We do **not** model wall-thickness ownership in MVP (FIG notes this is jurisdiction-specific and often undefined even in 2D drawings).

5. LADM research

5.1 What the standard is now

ISO 19152 LADM Edition I (2012) is superseded in parts by Edition II:

- **ISO 19152-1:2024** — generic conceptual model (Party, Admin, Spatial)
- **ISO 19152-2:2025** — land registration, including `LA_LegalSpaceBuildingUnit`, `LA_LegalSpaceUtilityNetworkElement`, `LA_LegalSpaceCivilEngineeringElement`, 2D/3D/mixed spatial units, survey sources including point clouds and GNSS
- Also published: valuation (part 4) and spatial plan (part 5)

LADM is explicitly a **descriptive conceptual model**, not a data-product specification. Full “LADM-conformant” claims require an abstract test suite we will not run in SIH.

5.2 Decision

Option	Verdict
Custom ad-hoc tables only	Reject — untranslatable to DoLR/ISO language
Strict LADM-conformant	Reject — overkill; we are not a national LA system
LADM-aligned	Accept — our tables map 1:1 onto LADM classes; we do not implement every association

Mapping

Our table	LADM concept
party	LA_Party
baunit	LA_BAUnit
rrr	LA_Right / LA_Restriction / LA_Responsibility
spatial_unit	LA_SpatialUnit + legal-space subtypes
building / floor	physical references (ExtPhysicalBuildingUnit analogue)
source_dataset / survey	LA_SpatialSource / LA_SurveySource
version columns	VersionedObject
ulpin_id	external identifier

Reason: judges from DoLR can be answered with “ISO 19152-2:2025 aligned, not a certified LADM product.” That is accurate.

6. CityGML / 3D city model research

CityGML 3.0 (OGC 20-010, 2021) is a **physical** 3D city model: Building, Construction, Tunnel, Relief, PointCloud, Transportation, etc. LOD 0–3. Utility networks are **not** a core 3.0 module; they live in Utility Network ADE.

CityGML does **not** model RRR. Linking CityGML buildings to LADM legal spaces is a known research pattern (e.g. TestbedLU IFC–CityGML–LADM FME workbenches; many FIG papers).

Decision

Layer	Format
Internal store	PostGIS 2D polygons + 3D solids + attributes (LADM-aligned)
Visualisation	Cesium entities / glTF / 3D Tiles (runtime)
Export (advanced)	CityGML 3.0 Building LOD1 + optional Relief; GeoJSON; glTF
Not used internally	Full CityGML database (3DCityDB) — too heavy for SIH

CityGML is an **interoperability export**, not the system of record.

7. Data sources (honest)

Official SIH dataset field is **empty**. Do not invent Indian government dumps.

7.1 What exists in India (and what we can actually get)

Source	What it is	SIH availability	Use
CartoDEM (NRSC/Bhuvan)	IRS DEM, ~1 arc-sec (~30 m); vertical accuracy ~8 m (90%)	Free download with daily tile limits	Terrain context only. Too coarse for floors.
SVAMITVA drone/LiDAR	High-res village abadi mapping	Scheme-held; not an open national LAS dump	Cite as production future source; not our demo file
SoI CORS	RTK/NRTK, RINEX; ITRF2008 epoch 2005 processing (independent of NSRF GCP network — flagged inconsistency in Current Science 2024/2025 papers)	Registration required; not needed live	CRS/accuracy metadata in the data model
TALD (IEEE Access 2025)	9 km² Thiruvananthapuram ALS, ~12 pts/m², UTM 43N, classified buildings/trees/ground	Paper exists; download not confirmed as frictionless	Optional if we obtain it legally; otherwise do not claim we have it
Official cadastral polygons	State Bhu-Naksha / DILRMP	Not openly bulk-downloadable for a random city block	Ingest format yes; live state GIS no
OSM buildings	Crowd footprints	Yes	Rough physical footprints; not legal parcels
Microsoft / Google building footprints	Derived polygons	Possible with licence check	Physical only

7.2 MVP dataset strategy (scientifically honest)

Primary demo dataset (we will author): demo/pune_kothrud_block/ — a **synthetic but georeferenced** 1-parcel / 1-building scene placed in a real Indian urban CRS (EPSG:32643, UTM 43N) with:

- 1 cadastral polygon (~800–1500 m²) with a **placeholder 14-char parent ULPIN** labelled OFFICIAL_ULPIN_PLACEHOLDER or ingested string.
- 1 G+4 or G+7 apartment building.
- Floor-plan GeoJSON for 2 representative floors (repeated with translation in Z).
- Synthetic airborne-style LAS (building + ground + trees) **or** a public non-Indian cloud used only for the extraction algorithm, clearly labelled.
- 1 underground water line + 1 sewer line as 3D polylines.
- 1 deliberately invalid overlapping unit for the validation demo.

Secondary algorithm benchmarks (labelled non-cadastral):

- ISPRS Vaihingen 3D / similar open ALS for building-extraction metrics.
- Public raster floor-plan samples for vectorisation experiments (not Indian legal plans).

Rejected false claims: “we will use SVAMITVA LiDAR for Pune”; “Bhuvan 30 m DEM gives floor heights”; “we have DoLR parcel APIs.”

7.3 CRS policy

Role	CRS
Storage / geometry engine	Projected metres, UTM zone of the site (demo: EPSG:32643) + ellipsoidal or orthometric Z documented per dataset

Role		CRS
Identifier / display		WGS84 lon/lat (EPSG:4326) for parent ULPIN compatibility
Cesium		WGS84 / ECEF via runtime transform
Heights		Store z_ref enum: ELLIPSOIDAL_WGS84 ORTHOMETRIC_EGM LOCAL_SITE

CORS/GNSS: we store reported horizontal/vertical accuracy and EPSG. We do **not** build a live NRTK client in MVP.

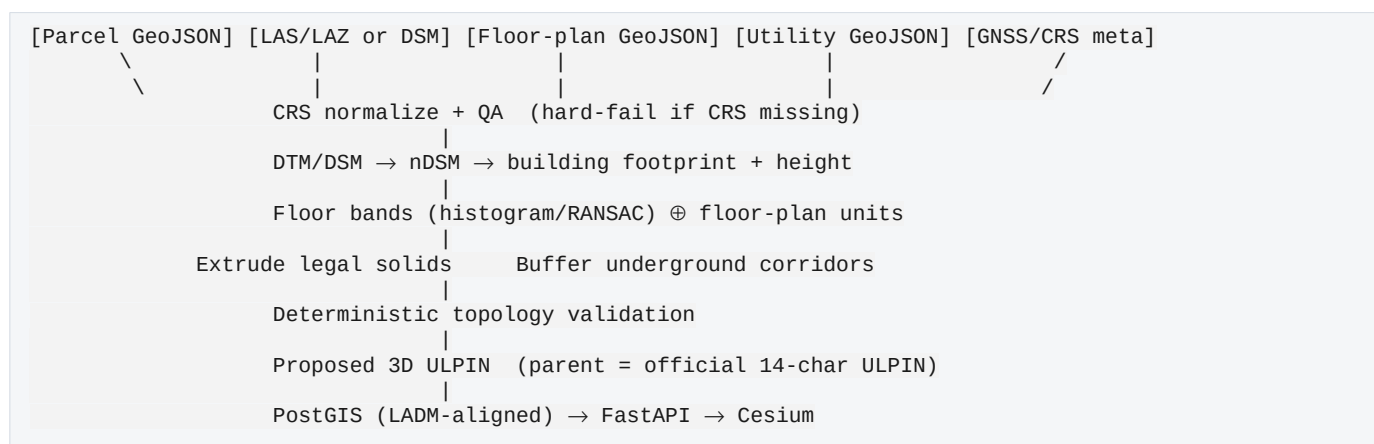
8. Data fusion pipeline

Default CRS: site UTM (metres). All stages log run_id, processor_version, confidence, and failure code.

#	Stage	Input	Output	Algorithm	Library	Accuracy target (MVP)	Failure	Fallback
0	Ingest	files + metadata	source_dataset, checksum, rows	dataset, MIME, CRS read	GDAL/PDAL	n/a	unreadable file	reject with error
1	CRS normalize	mixed EPSG	all in site CRS	pyproj / GDAL warp	GDAL, pyproj	reprojection error ■ survey error	missing CRS	hard fail (never guess silently)
2	Quality	clouds, rasters, polygons	QA report	density, nodata, ring validity	PDAL, Shapely	density ≥ 4 pts/m² preferred	empty/corrupt	continue with degraded confidence
3	Parcel align	cadastral polygon	spatial_union PARCEL	make valid, force 2D	Shapely/PostG	topology valid	self-intersect	ST_MakeValid + flag
4	DTM/DSM	LAS or rasters	DTM, DSM, nDSM	CSF or vendor classes; rasterize	PDAL CSF, rasterio	DTM ±0.3–1 m if dense ALS	no ground class	morphological filter / provided DTM
5	Building extract	nDSM + cloud	footprint polygon	height threshold + planarity + morph	PDAL, Open3D, SciPy	IoU ≥ 0.7 vs reference footprint	vegetation confusion	OSM/manual footprint
6	Height	building points / nDSM	z_ground, z_roof	percentiles (p5 ground, p95 roof)	NumPy	±0.5–1.5 m typical ALS	too few points	metadata storeys × 3.0 m
7	Floor segment	height + optional indoor cloud + plans	z bands per storey	Z-histogram peaks; RANSAC planes; plan priority	Open3D, NumPy	±0.3 m if plans; ±0.5–1.0 m if height-only	no peaks	equal slice H / N_storeys

#	Stage	Input	Output	Algorithm	Library	Accuracy target (MVP)	Failure	Fallback
8	Plan integrate	GeoJSON/IFC spaces	unit polygons in site CRS	Helmert/affine georeference	Shapely	RMSE_xy ≤ 0.5 m if GCPs	no plan	whole-floor as one UNIT + COMMON unknown
9	Vertical construct	footprint/unit poly + zmin/zmax	POLYHEDRALS Z solid	CG_Extrude / prism builder	PostGIS SFCGAL	closed solid	non-simple poly	simplify then extrude
10	Underground	3D lines + diameter	corridor solid	sweep/buffer in XY + Z slab or cylinder approx	Shapely 3D prism	schematic, not survey	missing Z	default cover depth 1.2–2.5 m flagged
11	Topology	all solids	validation_	see §15	PostGIS, SFCGAL, Shapely	100% rule coverage on demo	overlapping units	block “validated” status
12	3D ULPIN	valid spatial unit	display ID + UUID	see §11	Python	unique in DB	missing parent	parent = proposed surface key
13	Persist	objects	rows	SQLAlchemy	PostgreSQL	ACID	constraint fail	transaction rollback
14	Serve	DB	JSON + glTF	REST	FastAPI	< 2 s hot demo	—	—
15	Visualise	entities	globe	3D extrusion + underground translucency	CesiumJS	judge-readable	WebGL fail	2D MapLibre fallback

Pipeline diagram



9. AI / ML strategy

9.1 Building extraction comparison

Method	Accuracy potential	Train data	Cost	SIH fit	Verdict
CSF + nDSM + morph + planarity	Good on ALS $\geq$ ~8–12 pts/m ²	none	low	high	Primary
RANSAC roof planes (Open3D)	Good roofs, weak footprints alone	none	low	high	used inside primary
PointNet++ / KPConv / RandLA-Net	High on similar train domain	large labelled clouds	high GPU	poor for MVP	Advanced only
Image instance segmentation (buildings)	Needs nadir imagery + labels	high	medium	weak without Indian labels	rejected for MVP
Hybrid: classical propose, ML refine	best long-term	medium	medium	after MVP	future

Primary: unsupervised ALS workflow (CSF ground filter → DSM/DTM → nDSM height mask → vegetation rejection by planarity/TRI → contour regularisation). This is the family used by open tools such as LasBuildSeg and multiple ISPRS building-mapping papers. No training set required.

Fallback: user-supplied or OSM footprint + height from DSM percentiles or $\text{storeys} \times 3.0 \text{ m}$.

Rejected: training PointNet on TALD during SIH; calling a random Hugging Face model “cadastral AI.”

9.2 Floor segmentation

Indoor LiDAR literature (e.g. storey histogram + RANSAC floors/ceilings) assumes **interior** point clouds. Airborne LiDAR usually sees **roof and ground**, not slabs. Therefore:

MVP algorithm (hybrid, geometry-first):

1. If floor-plan / BIM spaces exist: they **win** for unit boundaries. Z comes from plan metadata or $i \times h_{\text{storey}} + z_{\text{ground}}$.
2. Else if interior cloud exists: Z-histogram (bin 5–10 cm) → peaks = slabs → RANSAC horizontal planes ($\text{distance_threshold} \approx 0.15\text{--}0.30 \text{ m}$).
3. Else (airborne only): n_{floors} from metadata **or** $\text{round}((z_{\text{roof}} - z_{\text{ground}}) / 3.0)$ with h_{storey} default 3.0 m (NBC-typical residential; **flagged as assumed**).
4. Geometry validator: floors must be monotonic in Z, $\text{gap} \in [2.4, 4.5] \text{ m}$ unless metadata says otherwise, no overlap.

ML is not the proposer in MVP. If we add it later, it may propose n_{floors} from facade images; geometry still validates.

9.3 Topology

100% deterministic. No classifier. “Intelligent” in the PS is implemented as **rule-based spatial intelligence** (overlaps, containment, CRS, Z, parent-child), which is what cadastral systems actually use (Queensland eSurvey validation, Shenzhen 3D topology checks).

9.4 Presentation rule

Every automated geometry row has $\text{geom_origin} \in \{\text{SURVEY}, \text{PLAN}, \text{AI_DERIVED}, \text{SYNTHETIC}, \text{MANUAL}\}$ and is **never** shown as “legal title.”

10. Geometry strategy

10.1 How a 2D polygon becomes a 3D parcel

Canonical construction:

```
legal_solid = Extrude(footprint_xy, zmin, zmax)
```

where footprint_xy is a simple polygon in site CRS and [zmin, zmax] is the vertical extent of that legal space.

10.2 Representation comparison

Rep	Pros	Cons	Use
Voxel	easy occupancy	huge, non-legal	reject
Triangle mesh only	pretty	hard CSG/validity	viz only
CityGML solid	standard physical	poor RRR	export
IFC IfcSpace	BIM legal spaces	IFC stack heavy	advanced ingest
Multipatch	ESRI-centric	not our DB	reject
POLYHEDRALSURFACE Z + SFCGAL solid	native PostGIS CSG, volume, 3D intersection	prism assumption	canonical legal geom
2D poly + zmin/zmax columns	trivial, fast 2.5D checks	not a true solid	construction params (always stored)

Decision: store **both** geom_2d + zmin/zmax (fast parent/overlap in 2.5D) **and** geom_3d solid (volume, 3D intersection, demo mesh). Most apartments in India are vertical prisms; that matches Queensland “building format” more than arbitrary B-Rep. Non-vertical walls (rare in MVP) deferred.

Underground utilities: approximate as **horizontal prism / stadium-buffer extrusion** along centerline at given depth and diameter. True cylinder sweep is advanced.

Libraries: Shapely 2.x for 2D; PostGIS 3.5+ CG_Extrude, CG_MakeSolid, CG_Volume, CG_3DIntersection (SFCGAL). Open3D for point-cloud planes, not for cadastral solids.

11. 3D ULPIN design

11.1 Three candidates

- A — Geometry-encoded NLI (ISO 8000-118 / ECCMA PNIU-style)**
Encode centroid + storey into ~14 chars.
Reject: point, not volume; unstable if entrance moves; collisions; DoLR ULPIN already uses 14 chars for **surface** parcels; underground poorly represented.
- B — Opaque UUID only**
Stable, easy.
Reject as display ID: judges/DoLR cannot see relationship to official ULPIN. Keep UUID as **internal PK only**.
- C — Hierarchical proposed identifier + UUID persistence (SELECTED)**

Internal PK:	ulpin3d_uuid (UUIDv7, immutable)
Display ID:	IN-ULPIN3D/{parent_ulpin}/{su_class}/{local_code}/v{nn}
Example:	IN-ULPIN3D/12AB34CD56EF78/UNIT/F05-U501/v01

Part	Rule
IN-ULPIN3D	namespace; marks proposed extension

Part	Rule
parent_ulpin	official 14-char if known; else P{12-char fingerprint of WGS84 centroid} labelled proposed
su_class	PARCEL BUILDING FLOOR UNIT COMMON PARKING AIR SUBSURFACE UTILITY
local_code	stable within parent, e.g. F05-U501, UTL-WTR-01
vnn	legal version, not geometry-tweak version

Geometry hash (sha256 of WKT) stored separately for change detection.

11.2 Lifecycle

Event	ID behaviour
Geometry snap / QA fix	UUID unchanged ; hash updates; version unchanged ; provenance note
Floor modification (legal)	new version vN+1; old row status=SUPERSEDED
Subdivision	parent UUID remains; children get new UUIDs; derived_from links
Merger	new UUID; sources linked; old SUPERSEDED
Redevelopment / demolition	status=EXTINGUISHED ; UUID never recycled
Duplicate prevention	unique (parent_ulpin, su_class, local_code, version) plus unique UUID

Uniqueness: DB constraints. **Persistence:** UUID never reused. **Stability:** display local_code changes only on legal events.

12. Database schema

PostgreSQL 16 + PostGIS 3.5 + postgis_sfcgal.

```
CREATE EXTENSION IF NOT EXISTS postgis;
CREATE EXTENSION IF NOT EXISTS postgis_sfcgal;
CREATE EXTENSION IF NOT EXISTS pgcrypto;

-- enumerations
CREATE TYPE su_class AS ENUM (
  'PARCEL', 'BUILDING', 'FLOOR', 'UNIT', 'COMMON', 'PARKING',
  'BALCONY', 'AIR', 'SUBSURFACE', 'UTILITY'
);
CREATE TYPE geom_origin AS ENUM (
  'SURVEY', 'PLAN', 'AI_DERIVED', 'SYNTHETIC', 'MANUAL'
);
CREATE TYPE topology_status AS ENUM (
  'PENDING', 'VALID', 'INVALID', 'DEGRADED'
);
CREATE TYPE rrr_type AS ENUM (
  'RIGHT', 'RESTRICTION', 'RESPONSIBILITY'
);
CREATE TYPE object_status AS ENUM (
  'ACTIVE', 'SUPERSEDED', 'EXTINGUISHED'
);

CREATE TABLE source_dataset (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  kind        text NOT NULL, -- las, geojson, geotiff, ifc, csv
```

```

filename      text NOT NULL,
checksum_sha256 text NOT NULL,
epsg          int,
z_ref         text,
acquired_at   timestampz,
ingested_at   timestampz NOT NULL DEFAULT now(),
meta          jsonb NOT NULL DEFAULT '{}'
);

CREATE TABLE survey (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  method      text, -- GNSS, CORS, photogrammetry, ALS, total_station, unknown
  h_rmse_m    numeric,
  v_rmse_m    numeric,
  source_id   uuid REFERENCES source_dataset(id)
);

CREATE TABLE party (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  party_type  text NOT NULL, -- person, org, association, authority
  name       text NOT NULL,
  ext_ref     text
);

CREATE TABLE baunit (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  name       text,
  uid        text UNIQUE
);

CREATE TABLE spatial_unit (
  id          uuid PRIMARY KEY, -- ulpin3d_uuid
  parent_id   uuid REFERENCES spatial_unit(id),
  parent_ulpin text, -- official 14-char or proposed surface key
  su_class    su_class NOT NULL,
  local_code  text NOT NULL,
  version     int NOT NULL DEFAULT 1,
  display_id  text NOT NULL UNIQUE, -- IN-ULPIN3D/...
  status      object_status NOT NULL DEFAULT 'ACTIVE',
  geom_2d     geometry(Polygon, 32643) NOT NULL,
  zmin        double precision NOT NULL,
  zmax        double precision NOT NULL,
  geom_3d     geometry(PolyhedralSurfaceZ, 32643),
  volume_m3   double precision,
  geom_origin geom_origin NOT NULL,
  confidence  numeric CHECK (confidence BETWEEN 0 AND 1),
  topology_status topology_status NOT NULL DEFAULT 'PENDING',
  geom_hash   text,
  baunit_id   uuid REFERENCES baunit(id),
  valid_from  timestampz NOT NULL DEFAULT now(),
  valid_to    timestampz,
  created_at  timestampz NOT NULL DEFAULT now(),
  CHECK (zmax > zmin),
  UNIQUE (parent_ulpin, su_class, local_code, version)
);

CREATE INDEX su_gix_2d ON spatial_unit USING GIST (geom_2d);
CREATE INDEX su_gix_3d ON spatial_unit USING GIST (geom_3d gist_geometry_ops_nd);
CREATE INDEX su_parent ON spatial_unit (parent_id);
CREATE INDEX su_class_idx ON spatial_unit (su_class);

CREATE TABLE building (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  spatial_unit_id uuid UNIQUE REFERENCES spatial_unit(id),
  storeys_above int,
  storeys_below int,
  z_ground    double precision,
  z_roof      double precision,
  extraction_method text
);

CREATE TABLE floor (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  building_id uuid REFERENCES building(id),
  spatial_unit_id uuid UNIQUE REFERENCES spatial_unit(id),
  level_index int NOT NULL,
  label       text

```

```

);

CREATE TABLE rrr (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  baunit_id   uuid NOT NULL REFERENCES baunit(id),
  party_id    uuid NOT NULL REFERENCES party(id),
  spatial_unit_id uuid REFERENCES spatial_unit(id),
  rrr_type    rrr_type NOT NULL,
  share       numeric, -- undivided share 0-1 for COMMON
  description  text,
  UNIQUE (baunit_id, party_id, spatial_unit_id, rrr_type)
);

CREATE TABLE validation_result (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  spatial_unit_id uuid REFERENCES spatial_unit(id),
  run_id      uuid NOT NULL,
  rule_code   text NOT NULL,
  passed      boolean NOT NULL,
  severity    text NOT NULL, -- ERROR, WARN
  detail      jsonb,
  created_at  timestampz NOT NULL DEFAULT now()
);

CREATE TABLE process_run (
  id          uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  stage       text NOT NULL,
  processor_ver text NOT NULL,
  source_ids  uuid[] ,
  metrics     jsonb,
  started_at  timestampz NOT NULL DEFAULT now(),
  finished_at timestampz
);

```

Versioning: legal change inserts a new `spatial_unit` row (new version or new UUID per §11.2); old row gets `status + valid_to`. No in-place overwrite of legal geometry without a provenance event.

13. API architecture

Base: `/api/v1` JSON. Idempotent GETs. Processing endpoints return `run_id` and poll via `/runs/{id}`.

Method	Path	Purpose
POST	<code>/datasets</code>	upload + CRS metadata
GET	<code>/datasets/{id}</code>	provenance
POST	<code>/sites</code>	create a processing site (CRS, bbox)
POST	<code>/sites/{id}/process</code>	run full pipeline or <code>?stage=</code>
GET	<code>/runs/{id}</code>	stage logs, metrics
GET	<code>/parcels/{id}</code>	surface parcel + children
GET	<code>/buildings/{id}</code>	envelope, storeys
GET	<code>/floors/{id}</code>	units on floor
GET	<code>/spatial-units/{id}</code>	full legal space + 3D ULPIN + confidence
GET	<code>/spatial-units/{id}/model.gltf</code>	mesh
GET	<code>/underground</code>	utility units in bbox

Method	Path	Purpose
POST	/validate	topology run
GET	/validation/{run_id}	rule results
POST	/ulpin3d/issue	issue proposed IDs for validated units
GET	/search?ulpin=	parent or 3D ID lookup
GET	/health	db/sfcgal

No public write of “official ULPIN.” Issuance always sets namespace=IN-ULPIN3D and status=PROPOSED.

14. Frontend architecture

Choice: CesiumJS + React (Vite) + a 2D MapLibre overlay for the parcel table.

Option	Globe + CRS	Underground	3D Tiles	SIH fit
CesiumJS	native WGS84	translucency + collision off	native	selected
Three.js	DIY globe	DIY	extra loaders	rejected as primary
deck.gl	good overlays	weaker cadastral globe	Tile3DLayer	optional layer later
MapLibre	excellent 2D	poor 3D cadastre	limited	2D companion

30-second judge UI (left → right):

1. 2D parcel highlighted on globe.
2. Point cloud / DSM fades in.
3. Building extrusion appears.
4. Click floor → isolate slab.
5. Click Flat 501 → cyan legal volume + ID IN-ULPIN3D/.../UNIT/F05-U501/V01.
6. Slider slices Z; globe becomes translucent; sewer corridor appears.
7. “Validate” paints a red overlapping fake unit; “Fix” removes it.

Panels: Property card (ULPIN, parent ULPIN, class, zmin/zmax, volume, origin, confidence, topology). Never use the word “title granted.”

15. Validation (deterministic)

Rule	Detects	Engine
CRS_DEFINED	missing/unknown EPSG	custom
Z_RANGE	$z_{max} \leq z_{min}$, insane heights	SQL CHECK + Python
SIMPLE_2D	self-intersecting rings	Shapely / GEOS
CLOSED_3D	non-solid / not closed	SFCGAL CG_IsSoLid / volume

Rule	Detects	Engine
PARENT_CONTAIN	unit footprint not in parcel	ST_CoveredBy (2D)
UNIT_OVERLAP	3D intersection volume > ϵ	CG_3DIntersection + CG_Volume
FLOOR_GAP	storey gaps outside tolerance	custom Z
FLOOR_OVERLAP	overlapping Z bands	custom Z
FLOATING	unit not stacked on building Z	custom
DUPLICATE_VOL	geom_hash collision different ID	SQL
DISCONNECTED	MultiPolygon parts without flag	Shapely
BLDG_PARCEL	building footprint outside lot	PostGIS
UTIL_SURFACE	utility intersecting UNIT volume	SFCGAL (WARN unless easement)
ID_UNIQUE	display_id / business key	unique indexes

ϵ for volumetric overlap: **0.01 m³** on demo solids (avoid numerical noise).

Invalid geometry **cannot** receive topology_status=VALID and should not be shown as a clean cadastral card in the happy-path demo.

16. Evaluation

What we can honestly measure in SIH

Metric	How	Realistic target on demo
Building footprint IoU	vs hand digitised roof	≥ 0.70 if using our synthetic/open ALS
Floor count accuracy	vs metadata	100% on happy path with plans
Storey elevation MAE	vs plan	≤ 0.3 m with plans; ≤ 1.0 m height-only
Overlap rule precision/recall	seeded errors	100% on injected overlaps
Horizontal RMSE of plan georef	GCPs	≤ 0.5 m
Pipeline time one building	wall clock	< 60 s excluding huge LAS
Peak RAM	one tile	< 8 GB

City-scale FPS and national RoR integration are **out of scope** as claims. We can discuss architecture scaling (tiling, 3D Tiles) without fake numbers.

Hausdorff distance: optional on footprints if we have a reference polygon; skip if we only have synthetic truth we ourselves created (still useful as regression test).

17. Benchmark / test cases

Case	Data	Must prove
N1 Normal	G+4 regular, complete plans, clean LAS	full path to 3D ULPIN
D1 Difficult	L-shaped building, noisy cloud, missing 1 floor plan	degraded confidence, not crash
U1 Underground	water + sewer vs basement	corridor solids + WARN if clash
F1 Fail CRS	shapefile without EPSG	hard fail, clear error
F2 Fail overlap	two units same volume	INVALID + red viz
F3 Fail LiDAR	parcel + plans only	height-from-metadata path
F4 Fail plans	parcel + LAS only	whole-floor units, COMMON unknown

Each pipeline stage has a pytest using these fixtures.

18. MVP vs advanced vs not-in-SIH

MVP (must work end-to-end)

- 1 Indian-CRS demo site, 1 parcel, 1 building, ≥ 4 storeys, ≥ 2 units/floor, 1 corridor COMMON, 1 parking, 1 underground utility.
- Ingest GeoJSON + LAS/LAZ or DSM + floor-plan GeoJSON.
- Classical building extraction **or** fallback footprint.
- Floor bands + extrusion to solids.
- Proposed 3D ULPIN with parent 14-char field.
- Topology validation + injected error demo.
- Cesium 2D/3D, floor isolate, Z-slice, underground toggle, provenance card.
- LADM-aligned PostGIS + FastAPI.

Advanced (time permitting)

- IFC `IfcSpace` ingest.
- CityGML LOD1 export.
- RandLA-Net optional building class (off by default).
- Facade-based storey count.
- 3D Tiles for larger tiles.
- Multiple buildings in one block.
- ISO 8000-118 NLI as secondary location index.

Do not attempt in SIH

- National ULPIN webservice reverse-engineering presented as official.
- Live CORS streaming.
- Conclusive titling / legally binding certificates.
- Full IndoorGML navigation graphs.
- Training foundation 3D models.
- Real SVAMITVA/DILRMP bulk data scraping.
- Wall-thickness legal boundary debates as a product feature.
- City-wide production cadastre.

19. Demo strategy

Scene: “Sinhagad Road, Pune — one redeveloped residential parcel” (synthetic data, real CRS). Spoken line: *Official ULPIN names the land. We name the volume.*

Script (≤ 4 minutes)

1. Show 2D parcel + 14-char parent ULPIN.
2. Load point cloud / DSM.
3. Run extract → footprint + LOD1 extrusion.
4. Detect floors (histogram + 3.0 m check) → overlay plans for floor 5.
5. Click Flat 501 → legal solid + `IN-ULPIN3D/.../UNIT/F05-U501/v01`.
6. Show COMMON corridor as **association undivided share**, not a fake private title.
7. Vertical slice.
8. Toggle underground water main (easement WARN if it clips basement).
9. Load corrupted duplicate unit → validation FAIL (red).
10. Reprocess → VALID cadastral card with confidence 0.86 PLAN+SURVEY, not “AI legal truth.”

20. Risks

#	Risk	Mitigation
1	No official Indian LAS/parcels	Synthetic georeferenced demo + honest labelling
2	Airborne LiDAR cannot see floors	Plans-first Z; height heuristics flagged
3	Judges think we invented official ULPIN	Banner: PROPOSED extension; parent field immutable
4	SFCGAL solids fail on dirty polygons	ST_MakeValid, simplify, store 2.5D params always
5	Deep-learning rabbit hole	Classical pipeline frozen as MVP
6	CRS silent error	Hard-fail missing EPSG
7	Cesium underground camera	translucency + disable collision; 2D fallback
8	Over-scoped data model	One site, one building
9	Legal overclaim	geom_origin + “not a title” copy
10	ID instability	UUID PK; version on legal events only

21. Judge questions (evidence-backed short answers)

Why can't existing ULPIN solve this?

DoLR: PNIU “spatially pointing to the **surface** of the parcel.” One ID for the lot cannot distinguish Flat 501 from the sewer under it.

Why 3D?

Vertical cities create overlapping RRRs on the same X,Y. FIG and ISO 19152-2 exist because 2D projection is ambiguous.

India NGP 2022 also asks for 3D subsurface mapping.

Why not just a building ID?

A building is physical. Cadastre needs **legal spaces** (units, common, easements) with parties and shares. LADM separates those.

How is 3D ULPIN related to official ULPIN?

It is a **proposed extension**. Display IDs nest under the official 14-character parent. We do not mint government ULPIN.

Who owns common areas?

RERA s.17 + state apartment acts: undivided proportionate share, typically to the association; not partitioned. We model COMMON + shared RRR.

Underground rights?

UTILITY / SUBSURFACE spatial units; default RRR type RESTRICTION/easement unless a party is assigned. MVP is geometric + data-model, not a new statute.

Accuracy?

Report measured demo metrics, not national cm claims. Plans path ~0.3 m Z; ALS footprint IoU target ≥ 0.7 ; 30 m CartoDEM is not used for floors.

LiDAR missing?

Fallback: footprint + storey metadata / equal slice. Confidence drops; origin $\neq$ SURVEY.

Floor plans missing?

Whole-floor prisms; unit subdivision skipped; status DEGRADED.

Validate AI?

AI/classical geometry never auto-VALID. Topology rules + human/plan source beat model scores.

Scale to a city?

Architecture (tiles, PostGIS, 3D Tiles) can; this prototype is one block. Do not claim city production.

Government GIS?

Ingest shapefile/GeoJSON + EPSG. Export GeoJSON/CityGML LOD1. No fake Bhu-Naksha login.

Standard?

LADM-aligned ISO 19152-2:2025; OGC CityGML export optional; ULPIN parent per DoLR.

Export geometry?

Yes: WKT/WKB, glTF, GeoJSON footprint, optional CityGML.

Building modified?

New version or new units; old UUID extinguished/superseded; parent ULPIN remains until parcel mutation.

Subdivision/merger?

New UUIDs, `derived_from`, versions; never recycle IDs.

Duplicate IDs?

Unique indexes on UUID and business key.

Legally usable immediately?

No. Prototype for DoLR exploration. Not a conclusive title system.

Prototype vs production?

Prototype: one site, proposed IDs, synthetic/open data. Production would need official ULPIN API, survey-grade control, statute, and operational GIS.

22. Technology decision matrix

Area	Options	Choice	Why	Rejected because
Point cloud IO/filters	PDAL vs Open3D-only	PDAL + Open3D	PDAL for LAS/LAZ/CSF pipelines; Open3D for RANSAC/normals	PDAL-only weak interactive geometry; Open3D-only weak production LAS pipelines
GIS DB	PostGIS vs Spatialite vs Mongo	PostGIS+SFCGAL	3D solids, SQL validation, government-familiar	Spatialite weak 3D CSG; document DBs poor topology
Viz	Cesium vs Three vs deck	CesiumJS	globe, underground, WGS84	Three = DIY geospatial; deck = overlay library
Legal geom	SFCGAL vs mesh vs voxel	SFCGAL solids + 2.5D params	volume & intersection	voxels illegal; mesh-only hard to validate
Building extract	classical vs DL vs hybrid	classical; DL later	no labels, explainable	DL needs TALD-scale training we may not have
Floors	RANSAC vs ML vs plans	plans ⊕ histogram/RANSAC	airborne LiDAR lacks slabs	ML storey counting unreliable without facades
Data model	custom vs aligned vs conformant	LADM-aligned	speak ISO without fake certification	full conformance untestable in SIH
City model	CityGML core vs ADE vs none	export LOD1 only	avoid 3DCityDB weight	forcing CityGML internally mixes physical/legal

23. What others are doing — and our gap

Already solved in literature/open source:

- ALS building footprints (many ISPRS pipelines; LasBuildSeg).
- Floor histogram + RANSAC (indoor reconstruction papers).
- LADM+Cesium apartment viewers (e.g. PointCloud3DLAS / Andinasari 2025: Rotterdam floor plans + AHN + PostgreSQL + Cesium).
- Netherlands BIM-to-3D-deed visualisation.
- Shenzhen 3D certificates.

Still weak / missing for **India + ULPIN**:

- Binding 3D units to **official 14-char ULPIN** as parent (almost nobody does this).
- Modelling **RERA/association common areas** rather than European condominium clones.
- Honest **no-LiDAR / no-plan** degradations with provenance.
- Deterministic topology as a first-class demo, not a hidden SQL check.
- Explicit **proposed vs official** identifier hygiene (other SIH teams are likely to mint a new “ULPIN” and overclaim).

Differentiators we will actually ship:

1. ULPIN-compatible hierarchical proposed 3D ID (parent never overwritten).
2. Legal-space vs physical-geometry split with Indian common-area RRR.
3. Prism solid engine + SFCGAL topology with injected-error demo.

- 4. Provenance/confidence on every volume.
- 5. Plans-first floor logic (correct for airborne Indian data realities).

24. Final decisions (no “both are possible”)

Architecture

Decision: Python FastAPI processing services + PostgreSQL/PostGIS/SFCGAL + React/Cesium client.
Reason: one language for PDAL/Open3D/Shapely; SQL for topology; Cesium for globe/underground.
Evidence: same pattern as recent 3D LAS academic prototypes; PostGIS SFCGAL docs for `CG_Extrude / CG_3DIntersection`.
Rejected: Django-monolith, Node geometry, 3DCityDB-first, Unity demo.
Consequence: team must install PostGIS with SFCGAL; Windows: use WSL or Postgres.app-equivalent installer carefully.

Data model

Decision: LADM-aligned schema in §12.
Rejected: full ISO conformance; purely custom GIS layers.

3D ULPIN

Decision: Candidate C — UUID + `IN-ULPIN3D/{parent}/{class}/{local}/vNN`.
Rejected: ISO 8000-118 as cadastral ID; UUID-only display.

AI/ML

Decision: classical extract + geometric floors; ML optional later. Topology not ML.

Geometry

Decision: 2D footprint + z-range + SFCGAL solid prisms.

Frontend

Decision: CesiumJS primary.

Standards

Decision: DoLR ULPIN as parent key; ISO 19152-2 aligned; CityGML export optional; OGC GeoJSON/gITF on the wire.

Datasets

Decision: authored synthetic Indian-CRS demo + optional public ALS for metrics; no fake SVAMITVA dump.

MVP / demo / metrics

As §§16–19.

25. Development order

Assume 14 focused days after this spec (adjust to team calendar).

Day	Outcome
1	Repo, Docker Compose (Postgres+PostGIS), CRS utilities, demo/ GeoJSON parcel
2	Schema migration, seed PARCEL + placeholder parent ULPIN
3	PDAL pipeline: read LAS, CSF, nDSM GeoTIFF

Day	Outcome
4	Building footprint extract + fallback OSM/manual
5	Height + floor-band detector + unit GeoJSON ingest
6	Extrude solids in PostGIS; volume; glTF export
7	Underground buffer solids
8	Topology rules + pytest fixtures (N1, F2, F1)
9	3D ULPIN issuer + API GETs
10	Cesium globe: parcel, building, click unit
11	Floor isolate, Z-slider, underground, validation colours
12	Provenance card, confidence, injected-error demo polish
13	Metrics notebook, README, judge Q&A dry run
14	Freeze demo data; record backup video; bug buffer

26. References

Tier 1 — Government / standards

- Smart India Hackathon, SIH26011 listing, <https://www.sih.gov.in/sih2026PS> (verified 18 Sep 2026).
- Department of Land Resources, *Bhu-Aadhar : Unique Land Parcel Identification Number (ULPIN)*, <https://dolr.gov.in/en/ulpin/> (updated 25 Apr 2024).
- Department of Land Resources, DILRMP / ULPIN guidelines PDF, <https://cdnbbsr.s3waas.gov.in/s3d69116f8b0140cdeb1f99a4d5096ffe4/uploads/2024/04/20240425160100930.pdf>.
- DoLR, DILRMP programme page, <https://dolr.gov.in/en/programmes-schemes/dilrmp-2/>.
- DST, *National Geospatial Policy 2022*, Gazette S.O. 6095(E), 28 Dec 2022, <https://dst.gov.in/sites/default/files/National%20Geospatial%20Policy.pdf>.
- Survey of India, CORS, <https://www.surveyofindia.gov.in/pages/continuously-operating-reference-stations-cors-and> <https://cors.surveyofindia.gov.in/>.
- NRSC, CartoDEM product policy, https://www.nrsc.gov.in/nrscnew/Dataproductions_Thematic_cartodem.php; Bhuvan FAQ (CartoDEM ~30 m, ~8 m vertical 90%).
- ISO 19152-1:2024; ISO 19152-2:2025 Land Administration Domain Model.
- ISO 8000-118:2025 Natural Location Identifiers (ECCMA lineage).
- OGC CityGML 3.0 Conceptual Model, OGC 20-010, <https://docs.ogc.org/is/20-010/20-010.html>.
- PostGIS SFCGAL: CG_Extrude, CG_MakeSolid, CG_Volume, CG_3DIntersection, <https://postgis.net/docs/>.
- The Real Estate (Regulation and Development) Act, 2016, s.17, India Code.
- Tamil Nadu Apartment Ownership Act, 2022; Karnataka Apartment Ownership Act, 1972; Punjab Apartment Ownership Act, 1995.

Tier 2 — Academic / FIG

- FIG, *Best Practices 3D Cadastres*, https://www.fig.net/resources/publications/figpub/FIG_3DCad/FIG_3DCad-final.pdf.

15. Kalogianni et al., LADM Edition II design including 3D, *Land Use Policy* / GDMC 2024, https://www.gdmc.nl/publications/2024/LUP_LADM_Ed2.pdf.
16. Stoter et al., Dutch 3D registration (Delft 2016), *ISPRS Int. J. Geo-Inf.* 2017, <https://doi.org/10.3390/ijgi6060158>.
17. Guo / Li / Ying et al., Shenzhen 3D cadastre practice (FIG / *Computers, Environment and Urban Systems* 2013).
18. Andinasari, *Point Cloud for 3D Land Administration System*, MSc TU Delft 2025, <https://www.gdmc.nl/publications/2025/MScThesisCitraAndinasari.pdf>.
19. Vijaywargiya & Ramiya, TALD, *IEEE Access* 2025, <https://doi.org/10.1109/ACCESS.2025.3546628>.
20. Storey histogram / RANSAC indoor segmentation: *Remote Sensing* 2018, <https://doi.org/10.3390/rs10081281>; *ISPRS Annals* 2024 X-4/W5-289.
21. Unsupervised ALS building mapping, arXiv:2205.14585 (not peer-reviewed; method family only).
22. Current Science papers on Indian geodetic / CORS vs NSRF inconsistency (2023–2025), e.g. <https://doi.org/10.18520/cs/v127/i2/147-152>.

Tier 3 — Implementations / datasets

23. Queensland Registrar of Titles Directions for the Preparation of Plans (Building vs Volumetric format), 2025.
24. Singapore Land Authority, Survey Maps and Plans (CP / CPST / subterranean RT).
25. CesiumJS documentation (globe translucency, 3D Tiles, vector draping).
26. Open3D `segment_plane`; PDAL CSF filter documentation.
27. 3DCityDB / py3dtilers (export path, not MVP core).

Tier 4 — Discovery only (not authority)

28. Community SIH mirrors (sih2026.vuce.in, GitHub scrapes of sih.gov.in). Used only to locate PS IDs; official text taken from sih.gov.in.
29. GitHub: citrandina/PointCloud3DLAS, MertcanErdem/LasBuildSeg, TestbedLU/Article-LADM-IFC-CityGML — implementation ideas only.

End of master specification. If a later official DoLR 3D ULPIN schema is published, replace §11 display format but keep UUID persistence and legal-space model.